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RESEARCH ARTICLE

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Imposing accurate wall boundary conditions in corrective-matrix-based moving particle semi-implicit method for free surface flow

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Summary

Corrective matrix that is derived to restore consistency of discretization schemes can significantly enhance accuracy for the inside particles in the Moving Particle Semi-implicit method. In this situation, the error due to free surface and wall boundaries becomes dominant. Based on the recent study on Neumann boundary condition (Matsunaga et al, CMAME, 2020), the corrective matrix schemes in MPS are generalized to straightforwardly and accurately impose Neumann boundary condition. However, the new schemes can still easily trigger instability at free surface because of the biased error caused by the incomplete/biased neighbor support. Therefore, the existing stable schemes based on virtual particles and conservative gradient models are applied to free surface and nearby particles to produce a stable transitional layer at free surface. The new corrective matrix schemes are only applied to the particles under the stable transitional layer for improving the wall boundary conditions. Three numerical examples of free surface flows demonstrate that the proposed method can help to reduce the pressure/velocity fluctuations and hence enhance accuracy further.

KEYWORDS

boundary condition, corrective matrix, free surface flow, MPS, particle method

1 | INTRODUCTION

The Lagrangian meshfree particle methods, including smoothed particle hydrodynamics (SPH)[1] and moving particle semi-implicit (MPS)[2] methods, have great advantages in simulating complicated interfacial flows. During recent years, the particle methods have been used to simulate free surface flow (eg, dam break,[2,3] sloshing,[4] waves,[5,6] and droplet deformation[7,8]), multiphase flow (eg, gas-liquid,[9-12] liquid–liquid[13-15] flows) and fluid–solid interaction flow (eg, fluid-rigid-[16-22] and fluid-elastic-structure[23,24] interactions). Meanwhile, the accuracy and stability of particle methods have been remarkably improved to enhance the reliability.

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Particularly, the particle shifting (PS) method[25] can stabilize simulations even if the coherent particle distributions would take place. After PS is applied, the high order formulations restoring discretization consistency (eg, moving least squares method,[26] least square moving particle semi-implicit (LSMPS) method,[27] kernel gradient correction in SPH,[28,29] and corrective matrix schemes in MPS[30-32]) can produce accurate and stable discretization for the inside particles that are not near boundaries. In this study, the corrective matrix schemes are adopted due to their simplicity. However, one problem is still remaining, namely, the boundary condition treatment methods are still similar to the original MPS.[2] In this situation, the error caused by boundaries (including both free surface and solid wall) will be dominant for the overall error. Particularly, as pointed out in the authors' recent work,[31] the overall error is not significantly reduced, even though the accuracy of discretization schemes is further improved for the internal particles by employing higher order corrective matrix. This indicates that the error caused by boundary conditions and stabilization techniques already plays a dominant role in the overall error after the first order corrective matrix is applied.[31] In these circumstances, enforcing stable and accurate high-order boundary conditions becomes a hot topic for particle method research.

The free surface boundary and wall boundary are two main physical boundaries for particle methods. Their recent improvements in terms of accuracy and stability are described below. Regarding free surface, it physically provides Dirichlet and Neumann (free-slip) boundary conditions for pressure and velocity, respectively. Numerically, the main difficulty comes from the lack of neighbor particles outside free surface. The original treatment method in MPS are as follows[2]: (1) free surface particles provide zero pressure in solving the pressure Poisson equation (PPE); (2) the same Laplacian, divergence and gradient models are applied to both free surface and internal particles; and (3) the particle collision model is utilized to guarantee the stability.[33] However, it is difficult to directly apply the high-order accurate schemes with corrective matrix to the free surface particles due to the incomplete neighbor support, as mentioned by Wang et al[34] and Duan et al.[31] Meanwhile, the virtual particle methods are proved to be helpful to improve accuracy and stability (eg, References 3,14,32,35-38), because virtual particles can compensate the neighbor particles outside free surface. Shibata et al[3] and Chen et al[35] derived the discretization models for free surface particles considering virtual particles. In these works, the boundary conditions are enforced on the virtual particles rather than free surface particles. Incidentally, the positions of virtual particles are not necessary according to References 3, 14, 35. More advanced methods to specify the positions of virtual particles were developed by Liu et al.[37,38] Besides, various methods by modifying and adjusting PS to enhance the stability at free surface were also developed by Lind et al,[39] Khayyer et al (optimized particle shifting, OPS),[40] Duan et al,[32] Oger et al,[41] Wang et al,[42] and Jandaghiana and Shakibaeinia.[43] Nevertheless, the general boundary condition treatment method of exact first-order accuracy is still missing for free surface in literature. The corresponding main difficulties will be explained in Section 3.2.2.

As for wall boundary, the physical conditions are Dirichlet (no-slip) and Neumann boundaries for velocity and pressure, respectively. For the former, the conventional method is to reflect the velocity of internal fluid particles to the mirror positions with normal to wall, as clearly illustrated by Cummins and Rudman[44] and Lee et al,[33] because the complete neighbor support is a key to guarantee accuracy. For the latter, the pressure of the wall/dummy particles can be extrapolated from the reference internal particle only (see fig. 4 by Lee et al[33]) or the closest internal particle (see fig. 1 by Lee et al[45]). Such methods can provide reliable results but are difficult to implement for the boundary with complex geometry (eg, corners or curves), especially in 3D. Meanwhile, based on the multiphase formulation developed by Shakibaeinia and Jin[46] and Duan et al,[9] it is possible to impose the no-slip (or free-slip) boundary conditions based on the multi-viscosity interaction models by regarding wall particles as fluid particles with infinite (or zero) viscosity. Such a method is simple and can be easily extended to complex geometry and implicit viscosity calculation.[47] Besides, a coupled dynamic solid boundary treatment algorithm with new least-squares-based interpolation scheme for estimating field functions of dummy wall particles was proposed by Liu et al[48] and further extended to multiphase flow by Chen et al.[12] Meanwhile, Takabatake et al[49] developed a heat flux model based on the divergence operator to impose the Neumann boundary condition for heat transfer. When corrective matrix (eg, References 31,32) is applied to improve accuracy near wall, the above methods can be straightforwardly employed by considering all the wall/dummy particles (see Figure 1) in calculating corrective matrix. In the above methods, the Neumann boundary condition is essentially converted to Dirichlet boundary conditions by approximating the field functions of dummy particles. Therefore, such boundary treatment method is not accurate enough, as mentioned by Tanaka et al.[50]

More recently, Tamai and Koshizuka[27] proposed constraint LSMPS schemes for imposing accurate Neumann boundary condition by considering boundary condition as a constraint for least squares methods. Tanaka et al[50] further applied

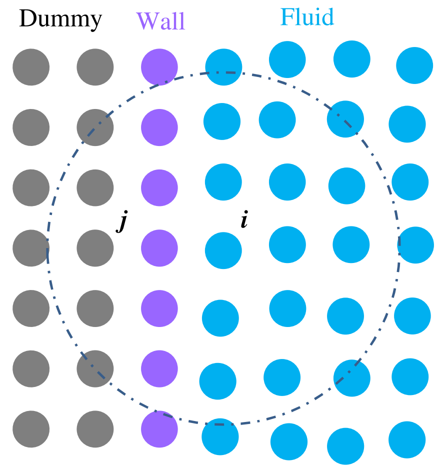


FIGURE 1 Particle arrangement near wall boundary [Color figure can be viewed at wileyonlinelibrary.com]

this method to the mesh wall boundary composed of imaginary particles along wall boundary. Improved formulations for wall Neumann boundary flux of pressure were also proposed in Reference 50. Very recently, Matsunaga et al[51] further improved Neumann boundary condition treatment method by estimating the boundary normal flux based on field functions of internal particles and enforcing such constraint of boundary flux in the least squares method for deriving new discretization schemes. In References 27, 50, the wall particles have to be considered as reference particles, while the wall particles are only considered as neighbor particles in Reference 51. Therefore, the computational cost of the method in Reference 51 is smaller than those in References 27, 50. An improved and simplified particle arrangement method on mesh wall was also proposed in Reference 51. It is noted that these methods in References 27,50,51 are all developed based on the LSMPS method. The aim of this work is to generalize the previous corrective matrix schemes to incorporate the accurate Neumann boundary condition treatment based on the similar idea to Reference 51. In such a manner, the boundary condition treatment will be coupled in the new corrective matrix schemes. Meanwhile, it is noted that the existing method in Reference 51 was only applied to closed flow without free surface. Actually, applying such high-order schemes to free surface is quite challenging because of the incomplete support of neighbor particles and the inaccurate detection of free surface, as discussed in Section 3.2.2. In this study, the previously developed stable strategy in Reference 32 will be applied to free surface and nearby particles to provide a stable transitional layer at free surface. The developed method aims at producing stable and accurate simulations for free surface flow.

The outline of the article is as follows. The existing corrective matrix schemes and boundary condition treatment are briefly described in Section 2. The proposed method based on the generalized corrective matrix schemes coupling boundary conditions will be presented in Section 3. Several numerical examples, including the hydrostatic pressure problem, dam break flow, and viscous spreading flow, are simulated to show the improvement compared to the previous method in Section 4. The concluding remarks are summarized in Section 5.

2 | EXISTING MPS METHODS WITH CORRECTIVE MATRIX

The mass and momentum conservation equations for incompressible flows in the Lagrangian framework read:

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{u} = 0, \tag{1}$$

and

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \frac{\mu}{\rho}\nabla^2 \boldsymbol{u} + \mathbf{g}, \tag{2}$$

where $\boldsymbol{u}$ is velocity; p is pressure; $\mathbf{g}$ is gravity; ρ and μ are density and dynamic viscosity, respectively.

2.1 | Corrective matrix discretization schemes in MPS

The MPS models with first and second order corrective matrix (SCM) for all discretization models were developed in References 32 and 31, 47, respectively. In this study, the schemes with the SCM in Reference 31 are adopted. Specifically, the following fitting function is used in deriving SCM in a two-dimensional system[31]:

$$\frac{\phi_{ij}}{r_{ij}} \approx \phi_x \frac{x_{ij}}{r_{ij}} + \phi_y \frac{y_{ij}}{r_{ij}} + \frac{l_0}{2}\phi_{xx}\frac{x_{ij}^2}{l_0 r_{ij}} + \frac{l_0}{2}\phi_{yy}\frac{y_{ij}^2}{l_0 r_{ij}} + l_0\phi_{xy}\frac{x_{ij}y_{ij}}{l_0 r_{ij}}, \tag{3}$$

where ϕ is a scalar variable; i is a reference particle; j is a neighbor particle; $\boldsymbol{r}_i = (x_i, y_i)$ is the position vector of particle i; $\phi_{ij} = \phi_j - \phi_i$; $x_{ij} = x_j - x_i$; $y_{ij} = y_j - y_i$; $\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i$; $r_{ij} = \|\boldsymbol{r}_{ij}\|$; ϕ_x is the first partial derivative with respect to x; ϕ_{xx} is the second partial derivative with respect to x and so on. For the convenience of applying least squares (LS) optimization, the fitting function is rewritten to the following general form:

$$d_{ij} = \boldsymbol{Q} \cdot \boldsymbol{P}, \tag{4}$$

where $d_{ij} = \phi_{ij}/r_{ij}$ is the directional difference; the unknow row vector $\boldsymbol{Q}$ and generalized direction column vector $\boldsymbol{P}$ are defined as follows:

$$\boldsymbol{Q} = \left[\phi_x \;\; \phi_y \;\; \frac{l_0}{2}\phi_{xx} \;\; \frac{l_0}{2}\phi_{yy} \;\; l_0\phi_{xy}\right], \tag{5}$$

and

$$\boldsymbol{P} = \left[\frac{x_{ij}}{r_{ij}} \;\; \frac{y_{ij}}{r_{ij}} \;\; \frac{x_{ij}^2}{l_0 r_{ij}} \;\; \frac{y_{ij}^2}{l_0 r_{ij}} \;\; \frac{x_{ij}y_{ij}}{l_0 r_{ij}}\right]^T = \left[P_1 \;\; P_2 \;\; P_3 \;\; P_4 \;\; P_5\right]^T. \tag{6}$$

It is noted that $\boldsymbol{P}$ is dimensionless and defined between a pair of particles i and j. The original hyperbolic weight function in Reference 2 is adopted. A constant particle number density (PND) n_0 can be calculated for the uniform particle configuration. Taking w_{ij}/n_0 (w_{ij} is the weight function between particles i and j) as the weight in LS, the unknown vector can be also approximated based on the general fitting function (Equation (4)), as derived in Reference 31.

Specifically, the SCM is calculated from:

$$\mathbf{C} = \begin{bmatrix} (P_1, P_1) & (P_1, P_2) & \cdots & (P_1, P_5) \\ (P_2, P_1) & (P_2, P_2) & \cdots & (P_2, P_5) \\ \vdots & \vdots & \ddots & \vdots \\ (P_5, P_1) & (P_5, P_2) & \cdots & (P_5, P_5) \end{bmatrix}^{-1}, \tag{7}$$

where (P_α, P_β) is an inner product of the components P_α and P_β by looping all the effective neighbor particles based on the weight of w_{ij}/n_0 as follows:

$$(P_\alpha, P_\beta) = \sum_{j \neq i} P_\alpha \cdot P_\beta \frac{w_{ij}}{n_0}. \tag{8}$$

Based on SCM, the high-order gradient, divergence and Laplacian models are still similar to the original ones.[31] Namely, the gradient model is as follows:

$$\langle \nabla \phi \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} d_{ij} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}, \tag{9}$$

where $\boldsymbol{C}_1$ and $\boldsymbol{C}_2$ are the first and second rows of matrix $\boldsymbol{C}$, respectively; $d_{ij} = \phi_{ij}/r_{ij}$. The dimensionless product of $[\mathbf{C}_1 \; \mathbf{C}_2]^T$ and $\boldsymbol{P}$ corresponds to the direction vector between two particles (ie, $\boldsymbol{r}_{ij}/r_{ij}$). Similarly, the divergence model based on SCM reads:

$$\langle \nabla \cdot \boldsymbol{u} \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \boldsymbol{d}_{ij} \cdot \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}. \tag{10}$$

where $\boldsymbol{d}_{ij} = (\boldsymbol{u}_j - \boldsymbol{u}_i)/r_{ij}$ is the directional difference of velocity vector. Finally, the Laplacian model becomes:

$$\langle \nabla^2 \phi \rangle_i = \frac{2}{n_0} \sum_{j \neq i} \left\{ w_{ij} d_{ij} \frac{[\boldsymbol{C}_3 + \boldsymbol{C}_4]\boldsymbol{P}}{l_0} \right\}, \tag{11}$$

where $\boldsymbol{C}_3$ and $\boldsymbol{C}_4$ are the third and fourth rows of $\boldsymbol{C}$, respectively; $d_{ij} = \phi_{ij}/r_{ij}$. Note that the product of $[\mathbf{C}_3 + \mathbf{C}_4]$ and $\mathbf{P}$ in Equation (11) is a dimensionless scalar. The detailed derivation can be found in Reference 31.

2.2 | Existing wall boundary condition treatment

The common particle arrangement near wall boundary in MPS is presented in Figure 1. Namely, one layer of wall particles to represent the real boundary and two/three layers of dummy particles to compensate the neighbor support of fluid particles. In the previous work,[31,32] both dummy and wall particles are considered in calculating corrective matrix. Therefore, it is necessary to specify the variable values for these wall and dummy particles based on boundary conditions.

For velocity, the interaction between fluid and wall can be regarded as fluid-solid interaction. In this situation, it is assumed that both the wall and dummy particles take zero velocity and infinity viscosity. The following interaction viscosity is used between fluid and wall/dummy particles (see eq. 25 of Reference 32):

$$\mu_{ij} = \frac{2\mu_i \cdot \mu_j}{\mu_i + \mu_j} = \frac{2\mu_i \cdot \infty}{\mu_i + \infty} = 2\mu_i, \tag{12}$$

where μ_i is the viscosity of a fluid particle and μ_j is the infinity viscosity of a wall or dummy particle. Such an imposing method is simple and can be straightforwardly applied to the complex geometry in 2D or 3D. On the other hand, the pressure of wall particles is solved from PPE. Meanwhile, the pressure of dummy particles is assumed equal to that of the reference fluid particles to impose the Neumann boundary condition with zero flux in both solving PPE and calculating pressure gradient. Besides, the effect of gravity on the hydrostatic pressure of dummy particles can also be considered, as described by Lee et al.[33]

3 | PROPOSED MPS METHOD

3.1 | New corrective matrix schemes coupling boundary conditions

The Dirichlet boundary condition can be straightforwardly considered in the existing corrective matrix schemes in Section 2.1. However, the coupling of Neumann boundary condition into corrective matrix is not straightforward. Recently, Matsunaga et al[51] proposed a new method to impose Neumann boundary condition for LSMPS. Based on similar ideas, such an imposing method is developed for the corrective matrix schemes.

First, the fitting function for imposing Neumann boundary condition is derived below. As illustrated in Figure 1, particle i is a reference fluid particle, and particle j is a wall neighbor particle on which Neumann boundary condition is imposed. The variable ϕ of particle j can be approximated from that of particle i based on Taylor series expansion:

$$\phi_j \approx \phi_i + x_{ij}\phi_x + y_{ij}\phi_y, \tag{13}$$

where the second and higher order terms are neglected. Regarding ϕ as the only variable and taking the gradient of ϕ on both side of Equation (13), the following relationship can be obtained:

$$\nabla \phi_j = \nabla \phi_i + x_{ij}\nabla(\phi_x) + y_{ij}\nabla(\phi_y), \tag{14}$$

where $\nabla\phi_j = \nabla\phi|_{\boldsymbol{r}_j}$ is the gradient of ϕ at particle j and so on. Let $\boldsymbol{n} = \begin{bmatrix} n_x & n_y \end{bmatrix}$ be the boundary normal vector. The boundary normal flux of ϕ at particle j can be calculated from:

$$\frac{\partial \phi_{\mathrm{j}}}{\partial n} = \boldsymbol{n} \cdot \nabla\phi_j = \boldsymbol{n} \cdot (\nabla\phi_i + x_{ij}\nabla(\phi_x) + y_{ij}\nabla(\phi_y)), \tag{15}$$

where $\frac{\partial \phi_j}{\partial n} = \left.\frac{\partial \phi}{\partial n}\right|_{\boldsymbol{r}_j}$. Then the target is to expand Equation (15) into a form similar to the original fitting function of Equation (3). Taking the following relationships of gradient vectors:

$$\begin{cases} \nabla\phi_i = \begin{bmatrix} \phi_x & \phi_y \end{bmatrix} \\ \nabla(\phi_x) = \begin{bmatrix} \phi_{xx} & \phi_{xy} \end{bmatrix} \\ \nabla(\phi_y) = \begin{bmatrix} \phi_{xy} & \phi_{yy} \end{bmatrix} \end{cases}, \tag{16}$$

into Equation (15), the following formulation can be obtained:

$$\frac{\partial \phi_{\mathrm{j}}}{\partial n} = \begin{bmatrix} n_x & n_y \end{bmatrix} \cdot \left(\begin{bmatrix} \phi_x & \phi_y \end{bmatrix} + x_{ij} \begin{bmatrix} \phi_{xx} & \phi_{xy} \end{bmatrix} + y_{ij} \begin{bmatrix} \phi_{xy} & \phi_{yy} \end{bmatrix} \right). \tag{17}$$

With simple mathematical manipulations, one will have:

$$\frac{\partial \phi_{\mathrm{j}}}{\partial n} = \begin{bmatrix} n_x & n_y \end{bmatrix} \cdot \begin{bmatrix} \phi_x + x_{ij}\phi_{xx} + y_{ij}\phi_{xy} & \phi_y + x_{ij}\phi_{xy} + y_{ij}\phi_{yy} \end{bmatrix}. \tag{18}$$

Next, the above Equation (18) can be further expanded to:

$$\frac{\partial \phi_{\mathrm{j}}}{\partial n} = n_x(\phi_x + x_{ij}\phi_{xx} + y_{ij}\phi_{xy}) + n_y(\phi_y + x_{ij}\phi_{xy} + y_{ij}\phi_{yy}). \tag{19}$$

Finally, Equation (19) is reformulated to the following form similar to Equation (3):

$$\frac{\partial \phi_{\mathrm{j}}}{\partial n} = \phi_x n_x + \phi_y n_y + \frac{l_0}{2}\phi_{xx}\left(\frac{2n_x x_{ij}}{l_0}\right) + \frac{l_0}{2}\phi_{yy}\left(\frac{2n_y y_{ij}}{l_0}\right) + l_0\phi_{xy}\left(\frac{n_x y_{ij} + n_y x_{ij}}{l_0}\right). \tag{20}$$

It is noted that the dimensions of Equations (20) and (3) are the same, and it is possible to apply LS optimization in a general manner. For the convenience of applying LS, Equation (20) is reformulated to the general form of Equation (4):

$$d_{ij} = \boldsymbol{Q} \cdot \boldsymbol{P}, \tag{21}$$

where $d_{ij} = \partial\phi_j/\partial n$ is the new directional difference; the unknow vector $\boldsymbol{Q}$ is kept the same as Equation (5) and the new generalized direction vector $\boldsymbol{P}$ is defined as:

$$\boldsymbol{P} = \begin{bmatrix} n_x & n_y & \frac{2n_x x_{ij}}{l_0} & \frac{2n_y y_{ij}}{l_0} & \frac{n_x y_{ij} + n_y x_{ij}}{l_0} \end{bmatrix}^T. \tag{22}$$

In terms of the general form (Equations (4) and (21)), the difference between the Neumann (Equation (20)) and normal (Equation (3)) fitting functions only lies in the calculation methods of directional difference (d_{ij}) and generalized direction vector ($\boldsymbol{P}$).

The above Neumann fitting function (Equation (20)) is only applied when the neighbor particle j is a Neumann boundary particle. The original normal fitting function (Equation (3)) should be employed when the neighbor particle j is an internal particle or a Dirichlet boundary particle. Specifically, for a reference fluid particle (eg, particle i in Figure 1), $\boldsymbol{P}$ will be calculated by Equation (22) and $d_{ij} = \partial\phi_j/\partial n$ when neighbor particle j is a Neumann boundary particle; $\boldsymbol{P}$ will be calculated by Equation (6) and $d_{ij} = \phi_{ij}/r_{ij}$ when neighbor particle j is an internal particle or a Dirichlet boundary particle. Note that the unknown vector $\boldsymbol{Q}$ is the same in the Neumann and normal fitting functions. Meanwhile, $\boldsymbol{Q}$ can be estimated by the LS optimization based on the same general form (ie, Equations (4) and (21)) of fitting function, as derived in Reference 31. Therefore, the specific implementation of $\boldsymbol{P}$ and d_{ij} does not influence the LS-based derivation of $\boldsymbol{Q}$. In other words, the new corrective matrix schemes can be obtained just by redefining $\boldsymbol{P}$ and d_{ij} in the old schemes based on the boundary condition type of particle j.

The newly developed corrective matrix schemes coupling Dirichlet and Neumann boundary conditions are summarized as follows. First, the generalized direction vector $\boldsymbol{P}$ is redefined as follow:

$$\boldsymbol{P} = \begin{cases} \begin{bmatrix} \frac{x_{ij}}{r_{ij}} & \frac{y_{ij}}{r_{ij}} & \frac{x_{ij}^2}{l_0 r_{ij}} & \frac{y_{ij}^2}{l_0 r_{ij}} & \frac{x_{ij} y_{ij}}{l_0 r_{ij}} \end{bmatrix}^T, & j \in \text{Internal or Dirichlet} \\ \begin{bmatrix} n_x & n_y & \frac{2 n_x x_{ij}}{l_0} & \frac{2 n_y y_{ij}}{l_0} & \frac{n_x y_{ij} + n_y x_{ij}}{l_0} \end{bmatrix}^T, & j \in \text{Neumann} \end{cases}, \tag{23}$$

where $j \in$ Internal or Dirichlet means that j is an internal particle or a Dirichlet boundary particle; $j \in$ Neumann means that j is an Neumann boundary particle. Second, based on the general form of $\boldsymbol{P}$, namely $\boldsymbol{P} = \begin{bmatrix} P_1 & P_2 & P_3 & P_4 & P_5 \end{bmatrix}^T$, the corrective matrix can be still calculated from Equations (7) and (8). Finally, the newly developed gradient becomes:

$$\langle \nabla \phi \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} d_{ij} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}, \tag{24}$$

where the directional difference is redefined as follows:

$$d_{ij} = \begin{cases} \frac{\phi_j - \phi_i}{r_{ij}}, & j \in \text{Internal or Dirichlet} \\ \frac{\partial \phi_j}{\partial n}, & j \in \text{Neumann} \end{cases}. \tag{25}$$

Compared to the existing model (Equation (9)), the difference only lies in calculation of d_{ij} and $\boldsymbol{P}$. Similarly, the new divergence model is as follows:

$$\langle \nabla \cdot \boldsymbol{u} \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \boldsymbol{d}_{ij} \cdot \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}. \tag{26}$$

where the directional difference vector is redefined as follows:

$$\boldsymbol{d}_{ij} = \begin{cases} \frac{\boldsymbol{u}_j - \boldsymbol{u}_i}{r_{ij}}, & j \in \text{Internal or Dirichlet} \\ \frac{\partial \boldsymbol{u}_j}{\partial n}, & j \in \text{Neumann} \end{cases}. \tag{27}$$

Accordingly, the Laplacian model is calculated as follows:

$$\langle \nabla^2 \phi \rangle_i = \frac{2}{n_0} \sum_{j \neq i} \left\{ w_{ij} d_{ij} \frac{[\boldsymbol{C}_3 + \boldsymbol{C}_4] \boldsymbol{P}}{l_0} \right\}, \tag{28}$$

where the directional difference (d_{ij}) is equal to that of gradient, as defined in Equation (25).

Finally, some notes are summarized for the new schemes. First, the boundary conditions are only imposed on the wall particles by neglecting the dummy particles in Figure 1, because the wall particles precisely represent the boundary. Specifically, in the proposed discretization schemes above, only the fluid particles are regarded as reference particles. Meanwhile, the wall particles are considered as neighbor particles of a reference particle, while the dummy particles are not considered as neighbor particles. Such a treatment method is different from the previous corrective matrix schemes in References 31, 32, 47, where (1) both fluid and wall particles should be regarded as reference particles and (2) both wall and dummy particles are considered as neighbor particles of a reference particle. Second, the new corrective matrix

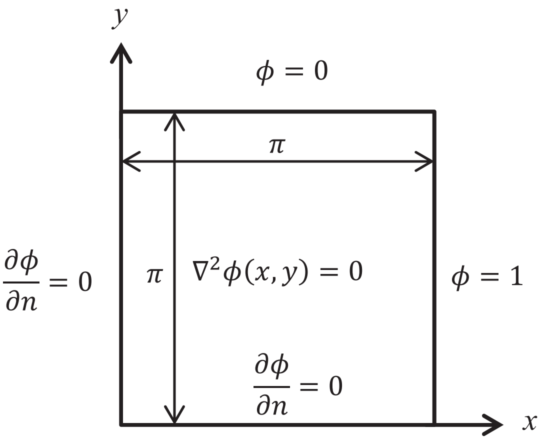


FIGURE 2 Sketch of the boundary value problem for verifying the new schemes

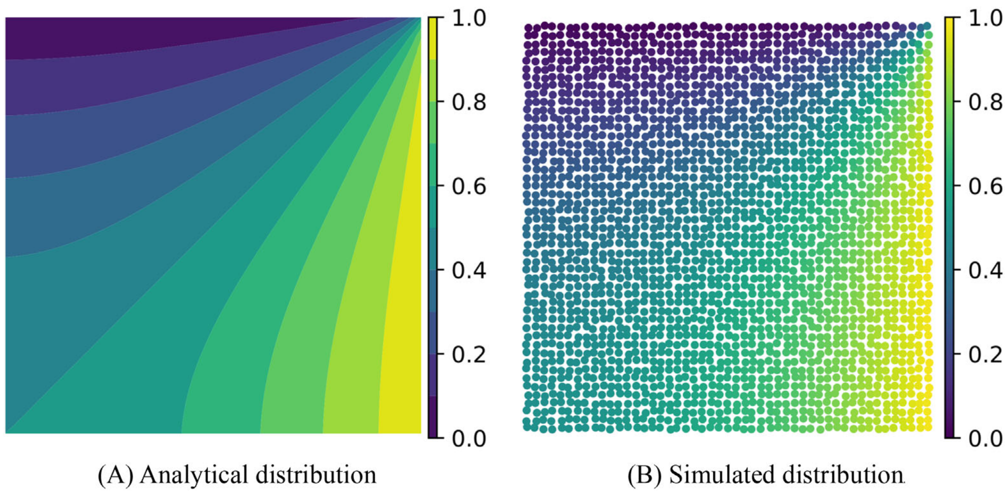


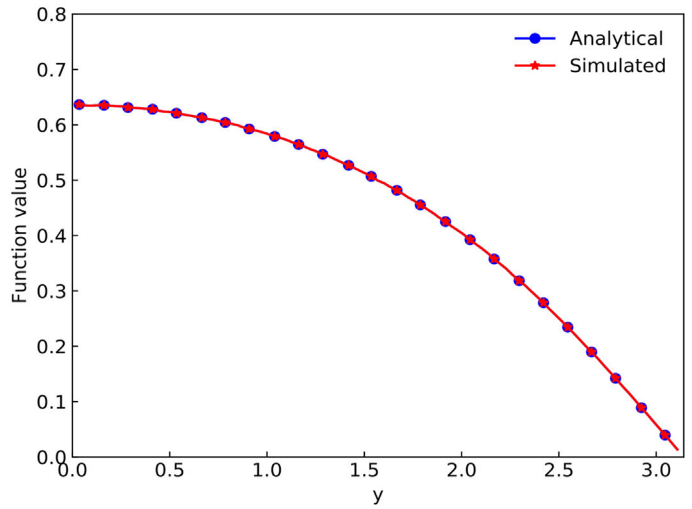
FIGURE 3 Comparison of analytical and simulated distributions for the boundary value problem using the resolution 50×50 [Color figure can be viewed at wileyonlinelibrary.com]

depends on the boundary conditions, while the existing one is free from boundary conditions. In this situation, the new corrective matrix for the particles near boundary must be updated after the boundary condition type is changed. In the existing schemes of References 31, 32, 47, corrective matrix is only calculated for one time regardless of the boundary condition type. Because the boundary conditions for velocity and pressure are different, some increase of computational cost is expected for the new schemes, which will be discussed later. Third, the new schemes can be regarded as extension/generalization of the existing ones to incorporate the Neumann boundary condition. In other words, if Dirichlet boundary condition is only considered or the Neumann boundary condition is conventionally considered in the Dirichlet manner (namely by estimating variable values for wall and dummy particles), the new schemes will be reduced to the existing ones automatically.

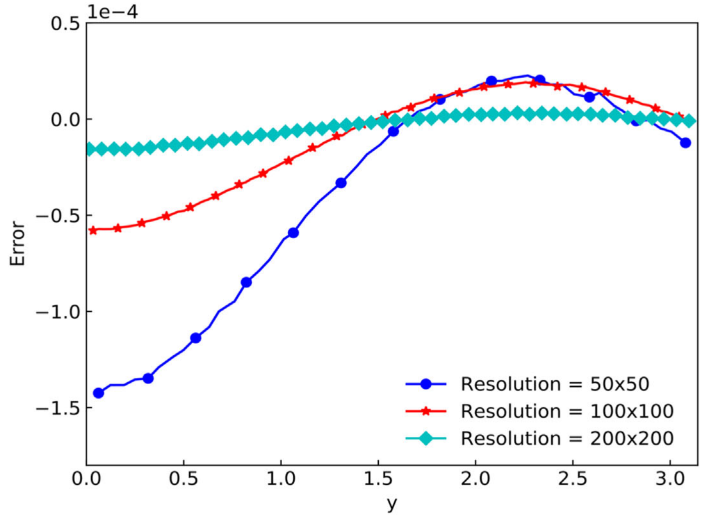
To verify the new corrective matrix schemes coupling boundary conditions, a boundary value problem is solved. The problem as well as its theoretical solution are described in sect. 4.2.1 of Reference 52. As shown in Figure 2, the problem is defined in a square domain of $(\pi \times \pi)$. The top and right boundaries are Dirichlet boundary conditions with values 0 and 1, respectively. The left and bottom boundaries are Neumann condition with zero flux. In this case, the right-top corner is a singular point. Due to this reason, the quantitative convergence order is not calculated, since the main purpose of this test case is to verify the developed schemes.

Three resolutions for internal domain, 50×50, 100×100, and 200×200, are used to solve the problem implicitly. Randomness is added to the particle positions, similar to Reference 47. The qualitative results are illustrated in Figure 3, where the simulated distribution of function values agrees reasonably with the analytical one. The quantitative results along the central vertical line ($x = \pi/2$) are further compared in Figure 4. The simulated profile of function values by the resolution 100×100 agrees exactly with the analytical result, as shown in Figure 4A. Meanwhile, the error decreases significantly after fine resolution is adopted, as shown in Figure 4B. These results verify the accuracy and convergence of the new schemes.

FIGURE 4 Distribution and error comparison along the central vertical line ($x = \pi/2$) [Color figure can be viewed at wileyonlinelibrary.com]



(A) Comparison of function value profiles (resolution 100×100)



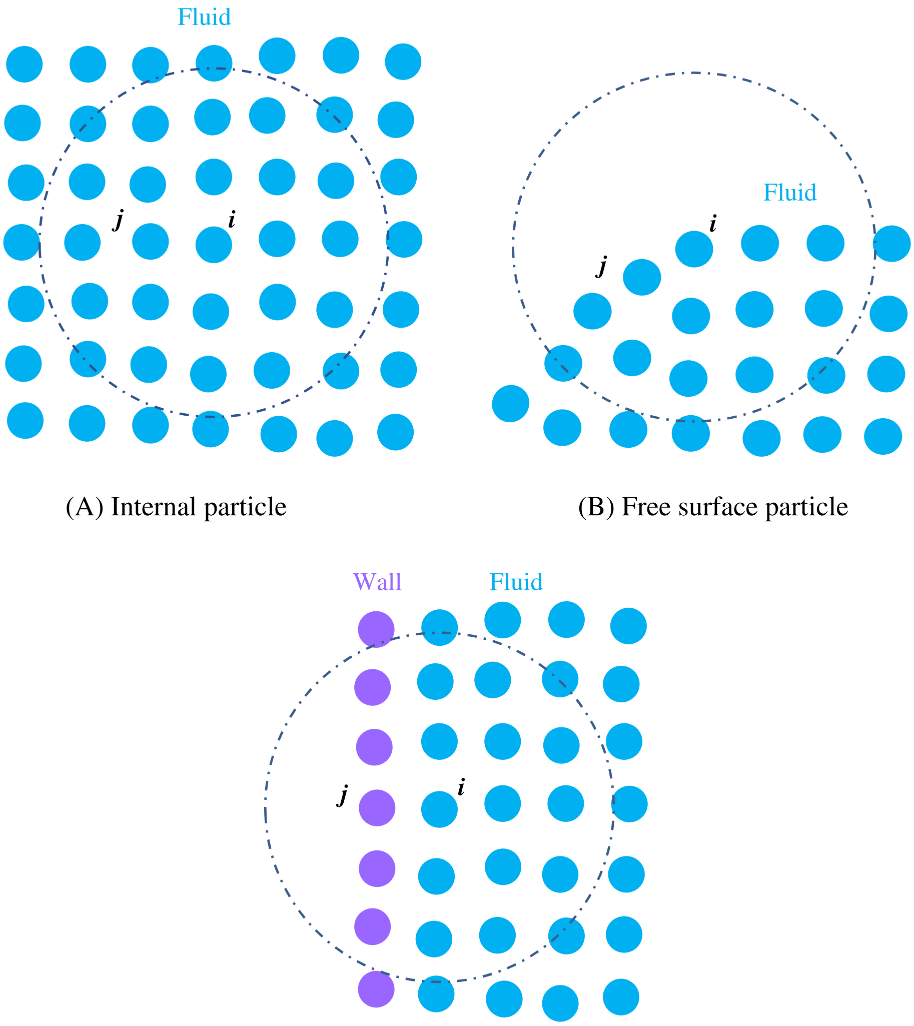
(B) Comparison of error profiles using different resolutions

3.2 | Free surface boundary condition

The above new schemes can be easily applied to wall boundary with Dirichlet velocity and Neumann pressure boundary conditions. However, when they are applied at free surface, unexpected instabilities can take place easily. This probably is the reason why the recent method with accurate Neumann boundary condition coupled for LSMPS is only applied to closed flows without free surface.[51] In this section, first, the reasons for the free surface instability are analyzed. Second, a stable transitional layer is applied at free surface.

3.2.1 | Error analysis at free surface

Numerical instability can be interpreted as the sharp increase/accumulation of error. In this perspective, the influence of neighbor particle distributions on error accumulation is investigated below to show how the incomplete support of neighbor particles can trigger the free surface instability.



(A) Internal particle

(B) Free surface particle

(C) Near wall boundary

FIGURE 5 Typical neighbor particle distributions for a reference fluid particle in the new corrective matrix schemes [Color figure can be viewed at wileyonlinelibrary.com]

First, the discretization models can suffer from biased error because of the incomplete/biased support of neighbor particles. The error terms seriously depend on neighbor particle distributions, as indicated by the discretization error analysis for the corrective matrix schemes in sect. 2.2 of Reference 31. The discretization error was derived by taking the dominant truncated error term of Taylor series expansion into the discretization schemes. Generally speaking, if the neighbor support is complete (eg, the internal case in Figure 5A), the error in every direction can cancel out or balance each other, so that the final error tends to be small and random. In this situation, error usually does not sharply accumulate over time to cause instability. On the other hand, if the neighbor support is incomplete and biased (eg, the free surface case in Figure 5B), the error information in some directions is missing. The final error cannot be well balanced and thus tends to be biased. Considering that the biased error may point to a specific direction, sharp accumulation or frequent fluctuation of biased error can take place over time more easily, compared to the random error. In this situation, the instability risk becomes high. For the reference particle near wall boundary (eg, the case in Figure 5C), even though only one layer of boundary particles (eg, wall particles) are considered as neighbor particles, these particles can significantly compensate the incomplete/biased neighbor support because of their large weight. In this situation, the biased error is remarkably suppressed, and instability risk is significantly reduced. It is noted that all the discretization models can suffer from biased error due to incomplete neighbor support in the new and old schemes. Since the pressure gradient model is prone to cause instability, its error will be analyzed below as an example to elaborate the above discussion.

Since Dirichlet boundary condition is imposed for pressure at free surface, the gradient model in the new and original schemes is the same in this situation. Therefore, error of the gradient model in Equation (9) is analyzed. After some time steps, error/fluctuation will occur in pressure. The error due to pressure fluctuation via the gradient model is analyzed below for free surface and nearby particles. Let Δp_i and Δp_j be the pressure fluctuations of particles i and j, respectively. Namely, the pressure of particle i and j can be split into:

$$\begin{cases} p_i = \overline{p}_i + \Delta p_i \\ p_j = \overline{p}_j + \Delta p_j \end{cases}, \tag{29}$$

where $\overline{p}_i$ and $\overline{p}_j$ are the true theoretical pressure of particle i and j, respectively. Taking Equations (29) into (9), the pressure gradient containing pressure fluctuation is as follows:

$$\langle \nabla p \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{(\overline{p}_j + \Delta p_j) - (\overline{p}_i + \Delta p_i)}{r_{ij}} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}. \tag{30}$$

Then, Equation (30) can be reformulated into

$$\langle \nabla p \rangle_i = \langle \nabla \overline{p} \rangle_i + \boldsymbol{E}_p, \tag{31}$$

where the true pressure gradient with second order accuracy, $\langle \nabla \overline{p} \rangle_i$, is as follows:

$$\langle \nabla \overline{p} \rangle_i = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{(\overline{p}_j - \overline{p}_i)}{r_{ij}} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}, \tag{32}$$

and the error *vector* due to pressure fluctuation, $\boldsymbol{E}_p$, is defined as follows:

$$\boldsymbol{E}_p = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{(\Delta p_j - \Delta p_i)}{r_{ij}} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}. \tag{33}$$

Next, Equation (33) can be further split into two terms as follows:

$$\boldsymbol{E}_p = \boldsymbol{E}_{p,j} + \boldsymbol{E}_{p,i}. \tag{34}$$

where the error due to Δp_j is:

$$\boldsymbol{E}_{p,j} = \frac{1}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{\Delta p_j}{r_{ij}} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}, \tag{35}$$

and the error due to Δp_i is:

$$\boldsymbol{E}_{p,i} = -\frac{\Delta p_i}{n_0} \sum_{j \neq i} \left\{ w_{ij} \frac{1}{r_{ij}} \left(\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P} \right) \right\}. \tag{36}$$

It is hard to determine whether $\boldsymbol{E}_{p,j}$ is biased or random at free surface, because it depends on the interaction between Δp_j and neighbor distributions. Due to the incomplete neighbor support, the possibility of $\boldsymbol{E}_{p,j}$ being biased is still high. On the other hand, it is clear that $\boldsymbol{E}_{p,i}$ is seriously biased at free surface but random and small for internal particles, which will be explained in next paragraph. When $\boldsymbol{E}_{p,i}$ becomes a dominant part of $\boldsymbol{E}_p$, instability can be triggered easily. When $\boldsymbol{E}_{p,i}$ is minor and $\boldsymbol{E}_{p,j}$ is not seriously biased (but random), instability will not happen. In other words, free surface instability does not definitely happen unconditionally but can happen accidently in some situations.

If the corrective matrix is not considered, the vector, $\begin{bmatrix} \boldsymbol{C}_1 \\ \boldsymbol{C}_2 \end{bmatrix} \boldsymbol{P}$, should be replaced by $\boldsymbol{r}_{ij}/r_{ij}$, as discussed in Section 2.2. In this situation, the summation in Equation (36) is well canceled out for internal particles (Figure 5A). Thus, $\boldsymbol{E}_{p,i}$ is small and basically random. On the other hand, as shown in Figure 5B, the summation in Equation (36) is not well canceled out for free surface and nearby particles due to the lack of neighbors. Thus, $\boldsymbol{E}_{p,\,i}$ is expected large and always in the direction of surface normal (may point to outside or inside of free surface), namely seriously biased. Because the corrective matrix $\boldsymbol{C}$ that only depends on the particle distribution is a constant matrix for particle i, it will accordingly correct all the direction vectors $(\boldsymbol{r}_{ij}/r_{ij})$. Therefore, the constant corrective matrix just resizes and rotates the error vector but does not change the biased nature of $\boldsymbol{E}_{p,\,i}$ near free surface. In brief, $\boldsymbol{E}_{p,\,i}$ is seriously biased near free surface but random and small for internal particles.

Meanwhile, the error vector $\boldsymbol{E}_p$ in Equation (33) can help to explain the instability mechanism due to closely approached particle pair for internal particles. When two particles are closely approached, their interaction coefficient is much larger than that of other neighbor particles. Even if $\Delta p_j - \Delta p_i$ is assumed as a random value, $\boldsymbol{E}_p$ will be seriously biased in the direction of this particle pair, easily triggering instability. Therefore, the unusual large particle interaction coefficient is prone to cause instability. For internal particle, PS has to be applied to avoid closely approached particle pair and guarantee stability.

Second, the reduction of neighbor particles can increase the error magnitude near free surface. For free surface and nearby particles, the diagonal elements of $\boldsymbol{C}^{-1}$ will be small due to the less neighbor particles, based on the calculation method in Equations (7) and (8). After the matrix inversion, the diagonal elements of $\boldsymbol{C}$ at free surface become larger than those of internal particles. Therefore, the interaction coefficients of p_{ij} in Equation (9) become larger at free surface due to corrective matrix. This can also be explained in a physical manner. When the interaction force in the gradient model is shared by less neighbor particles (eg, reducing interaction radius or near free surface), the interaction coefficients of p_{ij} in Equation (9) will become larger, and vice versa. Accordingly, the interaction coefficients can increase the magnitude of $\boldsymbol{E}_p$, as presented in Equation (33). Considering that the error tends to be biased at free surface, the enlarged $\boldsymbol{E}_p$ will further increase the instability risk.

Third, the fluctuation of free surface boundary can result in velocity/pressure fluctuations. Such fluctuation is mainly owing to free surface particle detection methods. As presented by Dilts[53] and Tamai and Koshizuka,[27] more accurate free surface detection methods are necessary when the high-order schemes based on least squares methods are employed. The reason is that the results of high-order schemes are sensitive to the boundary specified by free surface particles. Alternatively, this phenomenon can be explained in a physical manner. Imagine that the initial free surface boundary fluctuations exist in the hydrostatic water problem. Then, such fluctuations will indispensably result in small waves associated with velocity and pressure fluctuations. If such velocity and pressure fluctuations interact with biased error of the discretization models, the free surface instability probably happens easily, especially when the viscosity effect is negligible.

In summary, the seriously biased error caused by the incomplete/biased neighbor support, large interaction coefficients caused by the reduced number of neighbor particles via corrective matrix and fluctuation of free surface boundary caused by the surface particle detection methods, can significantly increase the instability risk near free surface.

3.2.2 | Stable transitional layer for free surface

After the high order schemes are employed, numerical instability can take place easily at free surface. Therefore, it is necessary to develop a stable transitional layer near free surface to overcome the instability issues. In this study, the stable free surface strategy developed in our previous work[32] is coupled with the new schemes. It is noted that other stable free surface methods can also be employed.

The basic ideas of stable transitional layer are that (a) the new schemes developed in Section 3.1 are only applied to internal particles and that (b) the original stable schemes with virtual particles are applied to free surface and nearby particles. First, the free surface and nearby particles are detected based on the approaches in sect. 2.2 of Reference 14. Specifically, the detection technique contains three steps. In Step (1), the free surface particles are roughly determined from the reduction of PND and neighbor particle number as follows:

$$n_i^* < 0.95 n_0, \tag{37}$$

and

$$N_i < 0.85 N_0, \tag{38}$$

where n_i^* is the temporary PND; N_i is the number of neighbor particles, and N_0 is the constant number of neighbor particles based on the initial regular particle distribution. If Equations (37) and (38) are not satisfied simultaneously, the reference particle will be regarded as an internal particle; otherwise, the free surface detection will be further considered based on the geometry conditions in Step (2), similar to Reference 27. Namely, if the following condition:

$$\begin{cases} \|\boldsymbol{r}_j - \boldsymbol{r}_i\| \geq \sqrt{2} l_0 \\ \|(\boldsymbol{r}_i + l_0 \boldsymbol{n}) - \boldsymbol{r}_j\| < l_0 \end{cases}, \tag{39}$$

or

$$\begin{cases} \|\boldsymbol{r}_j - \boldsymbol{r}_i\| < \sqrt{2} l_0 \\ \frac{\boldsymbol{r}_j - \boldsymbol{r}_i}{\|\boldsymbol{r}_j - \boldsymbol{r}_i\|} \cdot \boldsymbol{n} > \frac{1}{\sqrt{2}} \end{cases}, \tag{40}$$

is satisfied, the particle will be judged as an internal particle; otherwise, the particle will be finally judged as a free surface particle. In Equations (39) and (40), the unit normal vector $\boldsymbol{n}$ towards outside of free surface is computed from:

$$\boldsymbol{n} = \frac{\boldsymbol{N}}{\|\boldsymbol{N}\|}, \boldsymbol{N} = \frac{1}{n_{\mathrm{i}}} \sum_{j \neq i} \left\{ \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{r_{ij}} w_{ij} \right\}. \tag{41}$$

In Step (3), the free surface nearby particles are defined as the ones whose distance from the nearest free surface particle is smaller than $2.9 \times l_0$, which just slightly less than $r_e = 3.1 \times l_0$ in this study. More detection details can be found in Reference 14. Both the detected free surface and nearby particles play the role of the transitional layer.

Second, the stable schemes applied to free surface and nearby particles are described as follows. Virtual particles are supplemented to compensate neighbor support, and their positions are not necessary. The assumptions for virtual particles are as follows: (1) neighbor support is well compensated so that the corrective matrix is not necessary; (2) virtual particles take the same velocity with the reference particles (Neumann boundary condition); (3) their pressure is zero (Dirichlet boundary condition). With these assumptions, the Laplacian model for viscosity calculation is as follows:

$$\langle \nabla^2 \boldsymbol{u} \rangle_i = \frac{2d}{n_0 \lambda} \sum_{j \neq i} \{(\boldsymbol{u}_j - \boldsymbol{u}_i) w_{ij}\}, \tag{42}$$

where only the real fluid particles are considered as neighbor particle j; λ is the variance of particle distances defined in Reference 2; the above Neumann boundary condition of velocity is imposed. Similarly, the applied divergence model is as follow:

$$\langle \nabla \cdot \boldsymbol{u} \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ \frac{(\boldsymbol{u}_j - \boldsymbol{u}_i)}{r_{ij}} \cdot \frac{\boldsymbol{r}_{ij}}{r_{ij}} w_{ij} \right\}. \tag{43}$$

The Laplacian model for discretizing PPE is as follows:

$$\langle \nabla^2 p \rangle_i = \frac{2d}{n_0 \lambda} \sum_{j \neq i} \{(p_j - p_i) w_{ij}\} - \frac{2d}{n_0 \lambda} (n_i' - n_i^*) p_i, \tag{44}$$

where n_i' is the modified PND considering virtual particles (see Equation (45)); n_i^* is the temporary PND without considering virtual particles (see Reference 2); the above Dirichlet boundary condition of pressure is imposed. The modified PND (n_i') is calculated as follows:

$$\begin{cases} n_i' = \max(n_i^*, \widetilde{n}_i) \\ \widetilde{n}_i = n_0 + 0.1 \sum_{j \neq i}^{N_r} \{w_{ij} - w(l_0)\} \end{cases}, \tag{45}$$

where N_r is the number of real neighbor particles within the distance of l_0 from the reference particle. The conservative pressure gradient is employed for stability (similar to References 31,32):

$$\langle \nabla p \rangle_i = \frac{d}{n_0} \sum_{j \neq i} \left\{ \frac{(p_j + p_i)}{r_{ij}} \cdot \frac{\boldsymbol{r}_{ij}}{r_{ij}} w_{ij} \right\}. \tag{46}$$

The virtual particles can suppress the biased error by compensating the neighbor support. Meanwhile, the conservative model of Equation (46) also helps to cancel out the error near free surface because of the symmetric property of discretization. Therefore, the stability is significantly improved near free surface.

The applied stabilization methods are similar to those in Reference 31, 32. Namely, PS is applied to internal particles to keep particle distribution basically isotropic, and modified OPS is applied to free surface and nearby particles by carefully disabling the surface normal component of PS.[31,32] It is noted that the dummy particles in Figure 1 are employed differently from previous studies[31,32,47] to some extent. First, regarding the physical discretization terms (eg, gradient, divergence, and Laplacian models), the dummy particles are not considered as neighbor particles for corrective matrix or discretization models in this study (as aforementioned in Section 3.1) but considered in the previous studies,[31,32,47] due to the difference in treating boundary conditions. Second, regarding the stabilization terms, the dummy particles should always be considered as neighbor particles for the calculation of PS, OPS, and PND in both the present and previous[31,32,47] studies. The reason is that the dummy particles can effectively prevent the particle penetration into walls via the stabilization terms. It is also noted that the contribution of the area outside the wall meshes/polygons is considered only for stabilization terms (PS and PND) by an analytical integration method in Reference 51 (see details in sect. 2.5 of Reference 51). Such integral treatment corresponds to the contribution of dummy particles in this study.

3.3 | Overall algorithm

The velocity boundary condition can be classified into no-slip (Dirichlet) and free-slip (Neumann) boundary conditions. At the wall, only the former which is a physical boundary condition is considered in this study. The latter is difficult and not considered in this study because it is necessary to consider the boundary normal and tangent components of velocity separately. An accurate and consistent treatment method was presented in Reference 51. After the new schemes are applied to internal particles, the boundary condition of pressure is imposed differently from the existing methods.[31,32] The overall algorithm is slightly adjusted based on the authors' recent one that is designed for highly viscous flow.[47]

The overall algorithm and enforced boundary conditions are as follows.

(1) The temporary particle position is updated from:

$$\boldsymbol{r}^* = \boldsymbol{r}^k + \boldsymbol{u}^k \cdot \Delta t. \tag{47}$$

(2) The velocity divergence for PPE source is calculated from Equation (26) with wall particles taking zero velocity (Dirichlet boundary condition).

(3) The following PPE is solved:

$$\frac{1}{\rho} \langle \nabla^2 p \rangle_i^{k+1} = (1-\gamma) \frac{1}{\Delta t} \nabla \cdot \boldsymbol{u}^k - \gamma \frac{1}{\Delta t^2} \left\{ \frac{n' - n_0}{n_0} \right\} + \alpha \frac{1}{\Delta t^2} p_i^{k+1}. \tag{48}$$

where γ (0.001 by default) is the relaxation coefficient of mixing PPE source[54] and α (10^{-9} by default) is the artificial pressure coefficient.[55] The Neumann boundary condition of pressure for wall particles is determined as follows[51]:

$$\frac{\partial p_j}{\partial n} = \rho \boldsymbol{n}_j \cdot \boldsymbol{g}. \tag{49}$$

where $\boldsymbol{n}_j$ is the unit surface normal of a wall particle j. It is noted that the effect of gravity on pressure is considered by boundary condition in Equation (49) (like Reference 51) rather than by the PPE source in the existing MPS method.[2] Due to this reason, gravity is considered after solving PPE but before calculating pressure gradient as follows.

(4) A temporary velocity is calculated by considering gravity as follows:

$$\boldsymbol{u}^{*}=\boldsymbol{u}^{k}+\boldsymbol{g}\cdot\Delta t. \tag{50}$$

(5) Another temporary velocity is calculated by considering pressure gradient:

$$\boldsymbol{u}^{**}=\boldsymbol{u}^{*}-\frac{\Delta t}{\rho}\nabla p. \tag{51}$$

where the boundary condition of Equation (49) is also imposed.

(6) The final velocity is obtained by considering the viscosity term implicitly[47,56]:

$$\frac{u^{k+1}}{\Delta t}-\frac{\mu}{\rho}\nabla^{2}u^{k+1}=\frac{\boldsymbol{u}^{**}}{\Delta t}. \tag{52}$$

where the Dirichlet boundary condition (wall particles taking zero velocity) is imposed.

(7) The final particle position is updated by:

$$\boldsymbol{r}^{k+1}=\boldsymbol{r}^{k}+\frac{(\boldsymbol{u}^{k}+\boldsymbol{u}^{k+1})}{2}\Delta t. \tag{53}$$

In summary, compared the authors' previous methods in References 31, 32, the treatment of velocity and pressure boundary conditions for wall boundary is improved due to the adoption of the new schemes. Compared to Reference 51, the proposed method can simulate free surface flow stably. In the following section, the effects of the improved wall boundary are investigated compared to the existing method in Reference 47. More details on discretization schemes can be found in Reference 31. The solution algorithm with implicit viscosity calculation for highly viscous flow is adopted in both proposed and existing methods. The only difference between the proposed and existing methods lies in the treatment method of boundary condition.

4 | RESULTS AND DISCUSSIONS

In this section, three free surface simulation cases, namely hydrostatic pressure problem, dam break flow, and viscous spreading flow, are performed. The first two cases are mainly used to investigate the influence of improved pressure boundary of wall because of the negligible viscosity. The third case is used to examine the effect of improved velocity boundary of wall due to the high viscosity. The interaction radius $r_e = 3.1 \times l_0$ is adopted.

4.1 | Hydrostatic pressure

The first case is a hydrostatic pressure problem, which has been widely studied due to the issues of pressure oscillation and spurious velocity fluctuations. The former can be alleviated by applying the corrective matrix schemes (eg, References 31,32). The latter can be suppressed by considering a hydrostatic pressure to dummy particles based on gravity.[33] However, such a treatment method is not general for hydrodynamic flows and therefore not applied here. The performance of the new schemes in suppressing pressure and velocity fluctuations will be investigated below.

The adopted computational parameters are as follows. A water pitch with the size of 0.1 m × 0.1 m is simulated. The density is 1000 kg/m^3, and the viscosity is zero. The gravity is 10 m/s^2. Three average particle distances, $l_0 = 0.004$, 0.002, and 0.001 m, are adopted to simulate the hydrostatic flow for 5 seconds.

The pressure profiles of $l_0 = 0.002$ m at final moment ($t = 5$ seconds) are presented in Figure 6, where both methods can provide reliable, smooth, and similar pressure profiles. The quantitative errors of pressure are computed for further comparison. The dimensionless root-mean-square error (RMSE) of pressure at each moment is calculated as follows:

$$E_{\text{pressure}}=\frac{1}{\rho gH}\sqrt{\frac{1}{N}\sum_{i=1}^{N}(p_{i,\text{simu}}-p_{i,\text{theo}})^{2}}, \tag{54}$$

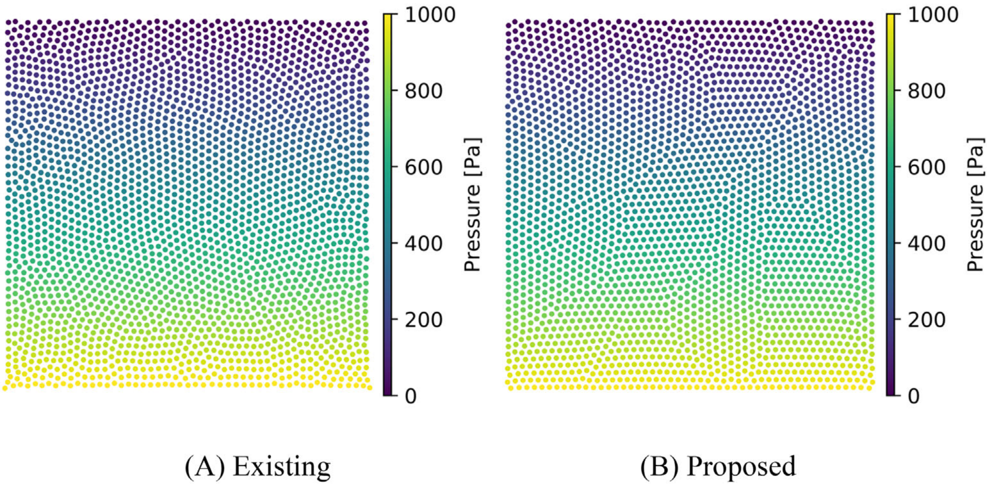


FIGURE 6 Comparison of pressure distributions simulated by the (A) existing and (B) proposed methods at $t = 5$ seconds [Color figure can be viewed at wileyonlinelibrary.com]

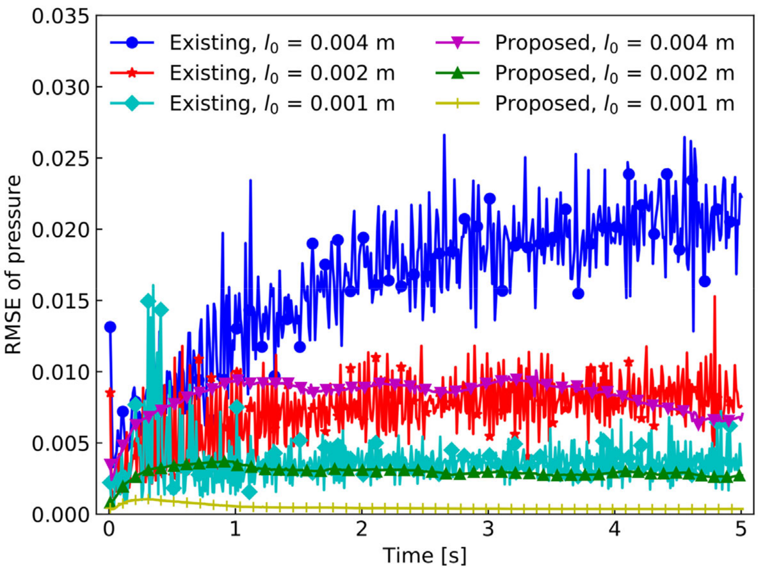


FIGURE 7 Comparison of dimensionless pressure errors simulated by the existing and proposed methods using different resolutions [Color figure can be viewed at wileyonlinelibrary.com]

where H is the height of free surface ($H = 0.1$ m) and N is the total number of fluid particles. The histories of such mean pressure error simulated by the two methods using various resolutions are contrasted in Figure 7. It can be observed that the proposed method remarkably suppresses the temporal oscillations of pressure and reduces the mean error, compared to the existing method. For both methods, the error decreases after l_0 is reduced, indicating convergence. The high-frequent pressure oscillations probably correspond to the velocity fluctuations in the flow, because velocity divergence provides the PPE source. The spurious velocity fluctuations simulated by the two methods at $t = 5$ seconds are compared in Figure 8A,B, where the fluctuation magnitude of the existing method is obviously larger than that of the proposed method. Meanwhile, in Figure 8A, the velocity fluctuations of the existing method are mainly produced near the vertical walls, because of the inaccurate treatment of pressure boundary condition. By contrast, in Figure 8B, the velocity fluctuations of the proposed method mainly occur near free surface, due to the stable transitional layer. It is noted that there are almost no velocity fluctuations for the internal particles in Figure 8B. In other words, the new boundary treatment method can significantly suppress the spurious velocity fluctuations.

In the proposed method, the source of velocity fluctuations is the stable transitional layer near free surface. It is necessary to investigate whether such velocity fluctuations spread downward to the internal area. Therefore, the simulations are performed for a long time (100 seconds) by both methods using $l_0 = 0.002$ m. The qualitative velocity distributions are presented in Figure 8C-F. In Figure 8C,E, there is almost no significant difference in the existing velocity distributions, indicating that the flow has reached a steady state. However, in Figure 8D,F, the region with velocity fluctuations increasing gradually, indicating the downward spreading of velocity fluctuations from free surface to internal area.

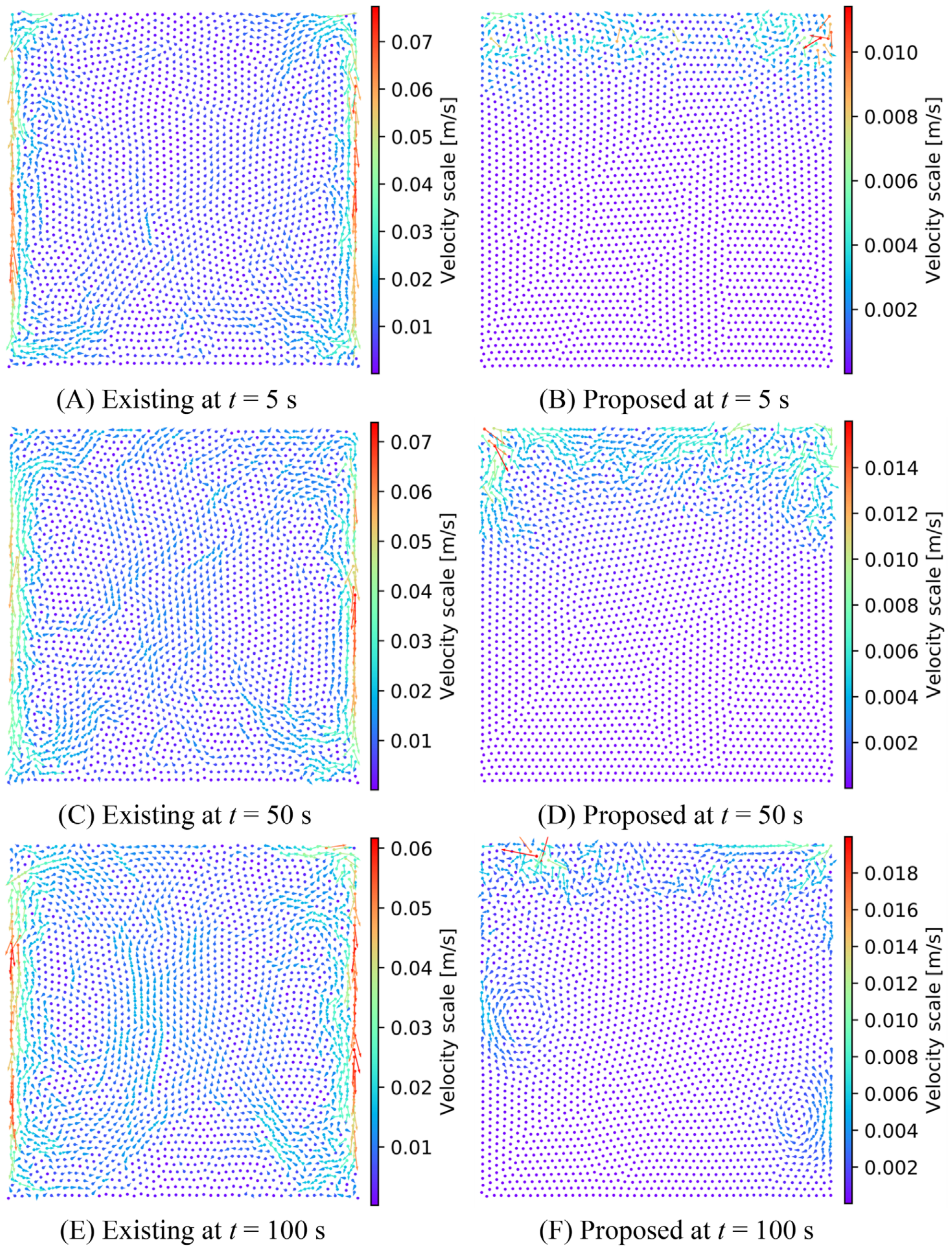


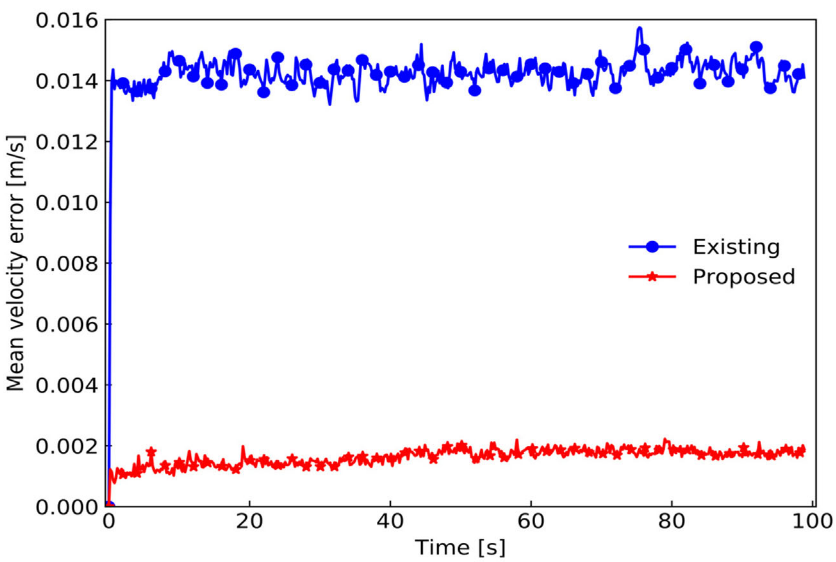
FIGURE 8 Comparison of velocity distributions simulated by the existing and proposed method at different moments [Color figure can be viewed at wileyonlinelibrary.com]

However it is noted that the velocity magnitude in Figure 8B,D,F by the proposed method is still much smaller than that in Figure 8A,C,E by the existing method.

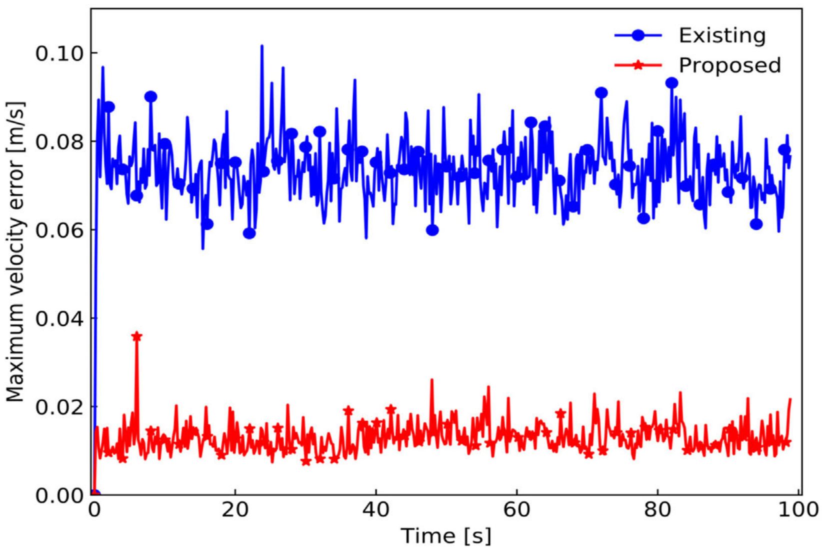
The velocity error is analyzed quantitatively to study the influence of downward spreading of velocity. Theoretically, velocity is zero everywhere in this problem, and the simulated velocity itself can be directly regarded as error. The mean velocity error can be calculated in the following RMSE manner at each moment:

$$E_u = \sqrt{\frac{\sum_{i=1}^{N}(u_i^2 + v_i^2)}{N}}, \tag{55}$$

where $\boldsymbol{u}_i = \begin{bmatrix} u_i & v_i \end{bmatrix}$. The maximum error of velocity magnitude can also be found at each moment. The mean and maximum velocity errors simulated by both methods are presented in Figure 9. Even though the present mean velocity error gradually increases in Figure 9A, the mean error is still much smaller than the existing one in 100 seconds. The reason is that the present source of velocity fluctuations is weaker and smaller than the existing ones. As shown in Figure 9B,



(A) Mean velocity error



(B) Maximum velocity error

FIGURE 9 Histories of mean and maximum velocity error over long time (100 seconds) [Color figure can be viewed at wileyonlinelibrary.com]

the present maximum error of velocity is obviously smaller than the existing one, indicating that the present fluctuation source is weaker than the existing one due to the improved boundary treatment. It is noted that the maximum velocity error does not increase over time in either method. Besides, the fluctuation source area in the present method (only near free surface) is obviously smaller than that in the existing method (near both free surface and vertical walls). Therefore, the present mean velocity error is expected still smaller than the original one even after longer time.

In summary, after the boundary condition treatments are improved in the new schemes, the errors and fluctuations of pressure/velocity are remark ably reduced, even though the velocity fluctuations caused by the stable transitional layer can spread gradually into internal region to some extent.

4.2 | Dam break

The second case is the dam break flow, where the simulation difficulty comes from pressure fluctuations and large deformation of free surface. The experiment case described in Reference 57 is simulated here in 2D. The initial flow

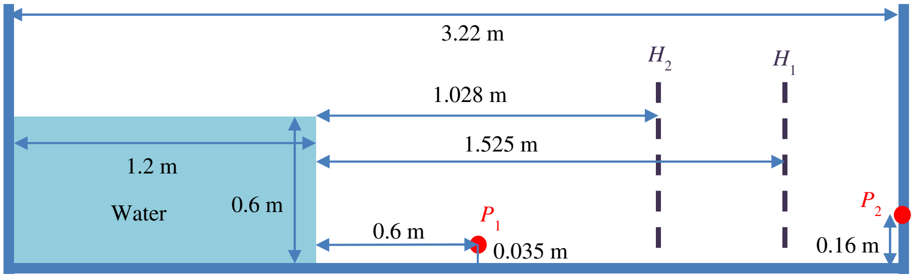


FIGURE 10 Sketch of the initial configuration and measurement points in the dam break problem [Color figure can be viewed at wileyonlinelibrary.com]

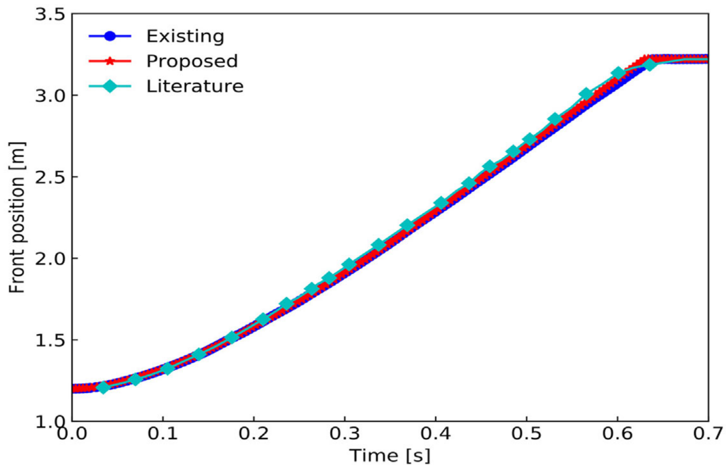
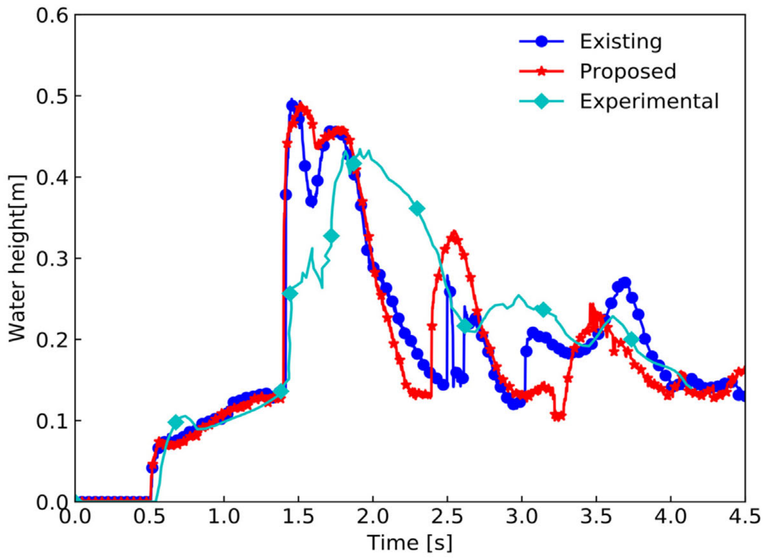


FIGURE 11 Comparison of front position histories of in dam break flow [Color figure can be viewed at wileyonlinelibrary.com]

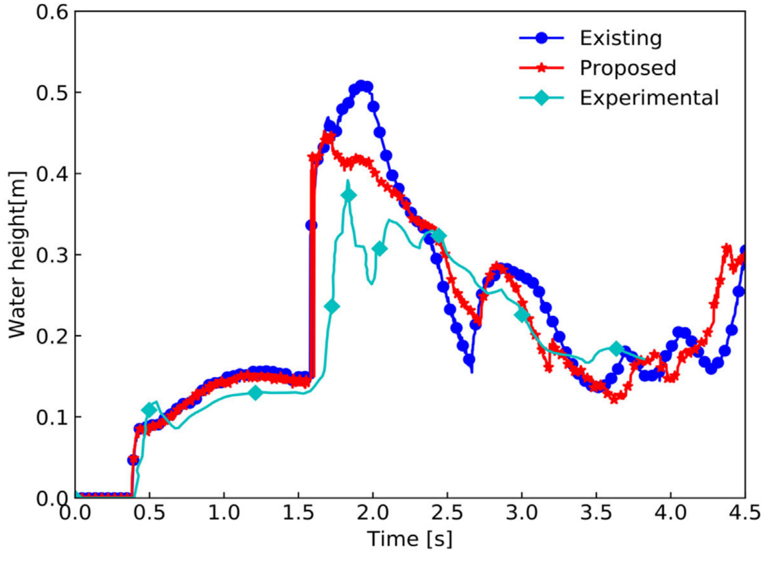
configuration and measurement points are sketched in Figure 10. Specifically, the initial water reservoir is placed in the left side of a tank. The histories of water depth along two vertical lines (H_1 and H_2) and pressure at point P_2 are measured in the experiment. The pressure at P_1 is only collected in simulations for contrasting the two methods. The density and viscosity of water in simulations are 1000 kg/m^3 and 0.001 Pa s, respectively. The gravity is 9.8 m/s^2. The particle size is 0.01 m and the largest allowable time step is 0.002 second.

First, the simulated results are compared with experiment ones for validation. The simulated histories of front positions are compared with literature results[37] in Figure 11, where good agreements are observed. The water depth histories are contrasted in Figure 12, where satisfactory agreements between simulations and experiment are observed. These quantitative results generally validate the two methods. However, it is difficult to determine which method is more accurate from these comparisons.

Second, the simulated pressure profiles are investigated to contrast the difference between two methods. The pressure distributions simulated by both methods at several typical moments are presented in Figure 13. Overall, the simulated pressure distributions by the two methods are similar to each other before $t = 2.0$ seconds. However, if the circled areas in Figure 13A,B are compared carefully, it is observed that pressure fluctuations still take place by the existing method while the pressure is smooth by the proposed method. Similar observations can also be found in the comparison between Figure 13C,D. Besides, the pressure at the right-bottom corner in Figure 13D by the proposed method is more smooth than that by the existing method in Figure 13C. At $t = 1.65$ seconds, the simulated pressure distributions by both methods are smooth and similar, as shown in Figure 13E,F. The quantitative pressure histories at P_1 and P_2 are presented in Figure 14. Note that measured data only at P_2 are available in experiment. Overall, the simulated pressure histories by both methods agree satisfactorily with experimental measurements, as shown in Figure 14A. It is noted that the first pressure peak is also not predicted in Reference 57. Before $t = 2.0$ seconds, there are almost no large pressure fluctuations, and the results by the proposed method exhibit less pressure fluctuations than those by the existing method. To demonstrate this phenomenon clearly, the simulated pressure histories at P_1 with high frequency output are contrasted in Figure 14B. It can be obviously observed that the proposed method significantly reduces the pressure fluctuations further compared to the existing method. Such an observation is consistent with the phenomena in Figure 13A-D. From $t = 1.65$-2.0 seconds, the entrapped void space in Figure 13E,F gradually disappears, and the corresponding reduction of free surface area results in significant pressure (and associated velocity) fluctuations, as circled in Figure 13G,H. After that, the coalescence



(A) Along the H_1 line



(B) Along the H_2 line

FIGURE 12 Comparison of water depth histories along two vertical lines of H_1 and H_2 [Color figure can be viewed at wileyonlinelibrary.com]

of fragmented free surface just begins to take place. Considering the instability nature of such free surface flow, free surface deforms violently and becomes quite unstable, as shown in Figure 13I-L. Negative pressure is also observed for internal flow due to vortices near the right-bottom corner. Because of the severely unstable free surface, large pressure fluctuations in both simulations take place after $t = 2.0$ seconds in Figure 14A. The turbulent dissipation terms probably help to significantly suppress the pressure fluctuations in such violent flow but are not considered in this article. Briefly, after $t = 2.0$ seconds, the improvements of boundary condition are obscured by the large pressure oscillations due to the severe coalescence and deformation of free surface.

The simulations are performed on a workstation with Intel Core i9-9960X (16 cores, 4.4 GHz) and GCC compiler (version 8.3). The computational time of the existing and proposed methods are 839.5 seconds and 956.2 seconds, respectively. Briefly, 13.9% of computational time is increased due to updating corrective matrix near wall boundary.

In summary, the simulated results of front position, water depth, and pressure agree satisfactorily with experimental measurements, validating the proposed and existing methods. Meanwhile, the proposed method can help to further suppress the pressure fluctuations compared to the existing method (see Figures 13A-D and 14B).

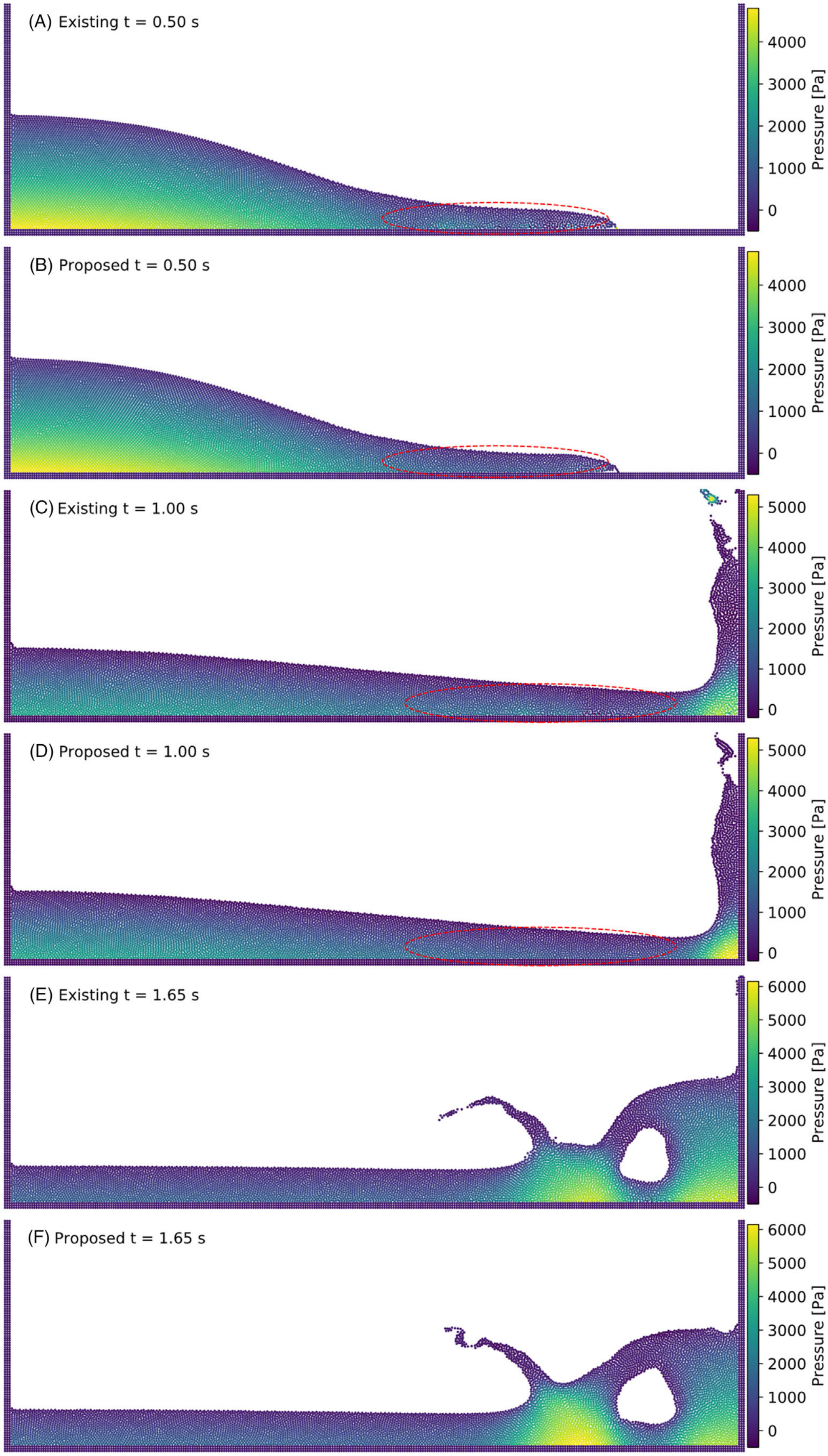


FIGURE 13 Comparison of simulated pressure distributions by the existing and proposed methods at different moments in dam break flow

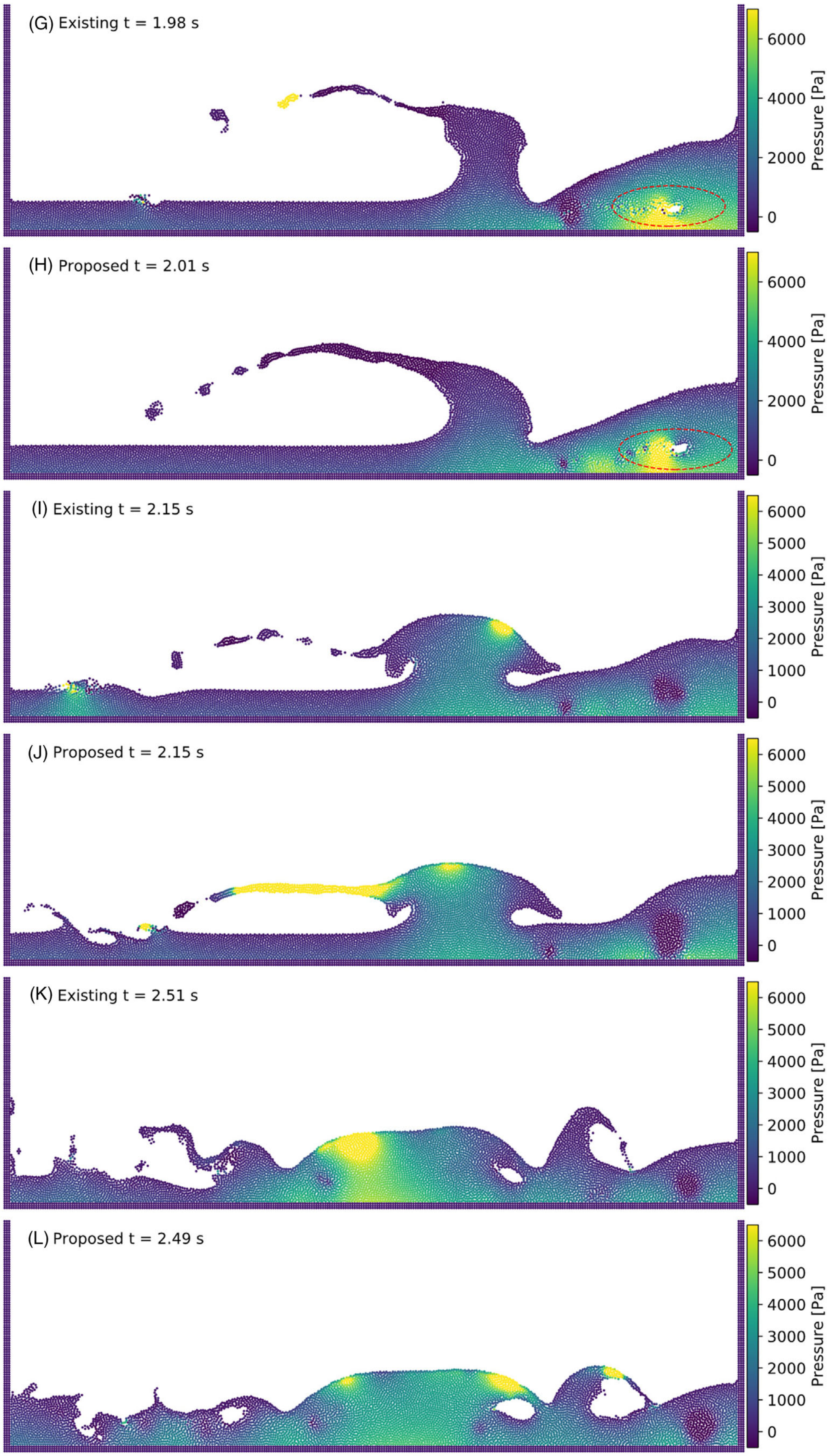
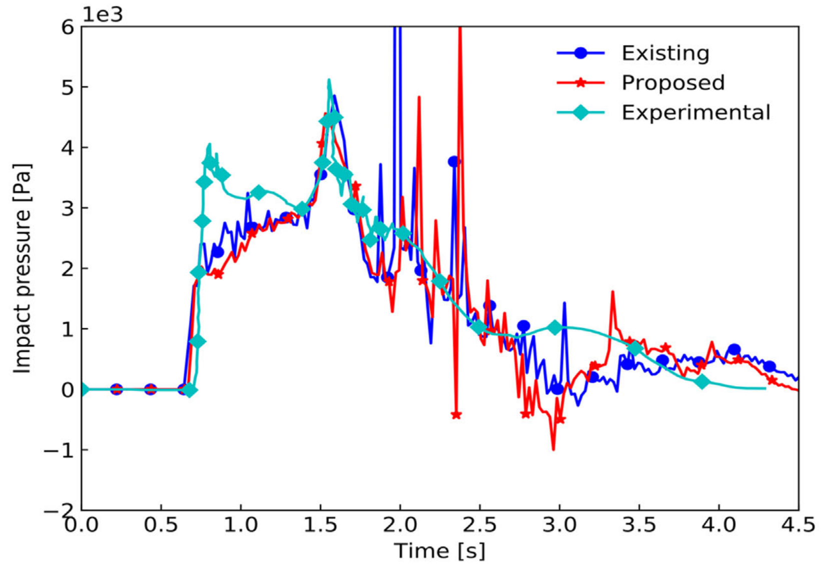
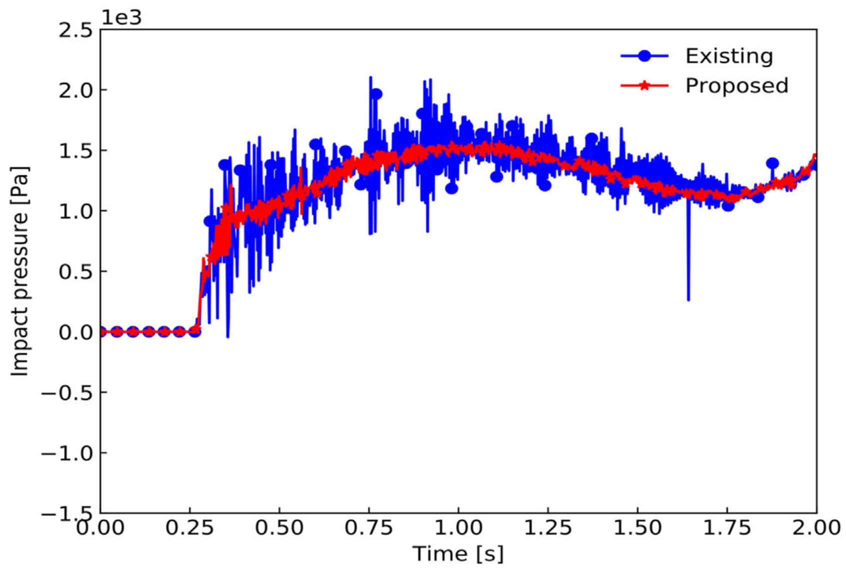


FIGURE 13 (Continued)

FIGURE 14 Comparison of pressure histories at two positions of P_2 and P_1 [Color figure can be viewed at wileyonlinelibrary.com]



(A) At point P_2



(B) At point P_1

4.3 | Viscous spreading flow

The third test case is the viscous spreading flow, in which the viscosity forces and velocity boundary condition will play the dominant role. Therefore, the influence of the improved velocity boundary condition at wall can be investigated.

The flow model is sketched in Figure 15, where the inflow boundary with given velocity is arranged at the left side of the spreading substrate. Fluid particles are not initially arranged but gradually injected from the inlet with a constant velocity of 0.075 m/s, until the end of simulations. The detailed implementation of inflow boundary can be found in sect. 2.3 of Reference 58. The gravity is 10 m/s^2. The density of injected fluid is 1000 kg/m^3. The dynamic viscosity is 10 Pa s, which is high so that the spreading flow is in the gravity-viscosity regime. Therefore, the following theoretical solution can be used for this problem[59]:

$$x_f(t) = \left(\frac{g}{3\nu}(q)^3\right)^{\frac{1}{5}} t^{\frac{4}{5}}, \tag{56}$$

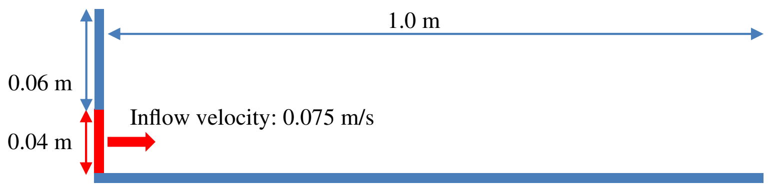


FIGURE 15 Sketch for the viscous spreading flow [Color figure can be viewed at wileyonlinelibrary.com]

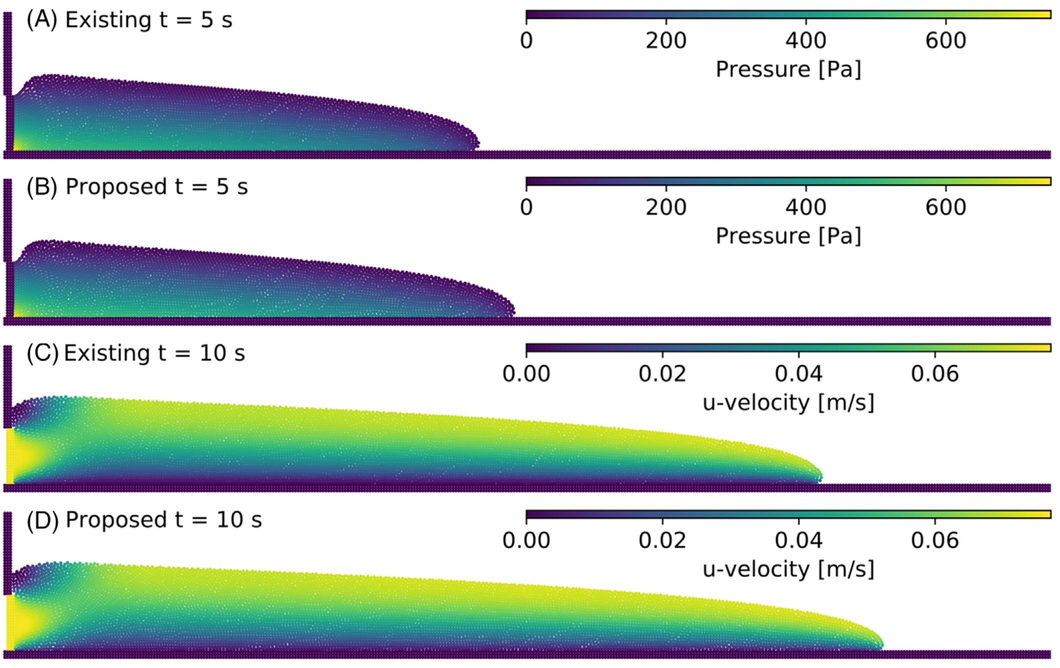


FIGURE 16 Comparisons of pressure ((A) and (B)) and u-velocity ((C) and (D)) distributions at two typical moments (5 and 10 seconds) [Color figure can be viewed at wileyonlinelibrary.com]

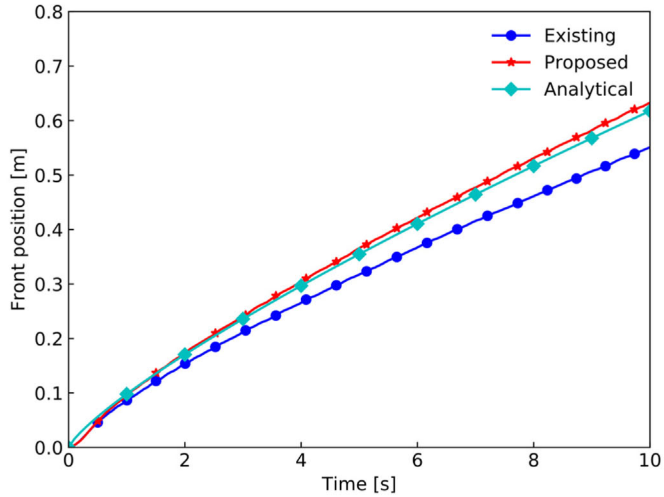
where t is time; x_f is the front position; g is gravity ($g = 10\,\mathrm{m/s^2}$); ν is kinematic viscosity ($\nu = 0.01\,\mathrm{m^2/s}$); q is the flow rate in 2D ($q = h_{\mathrm{inlet}} \times u_{\mathrm{inlet}} = 0.003\,\mathrm{m^2/s}$, where h_{inlet} is the inlet height and u_{inlet} is the inlet velocity). The flow is simulated for 10 seconds.

Three particle sizes, $l_0 = 0.004$, 0.002, and 0.001 m, are adopted in the simulations. The simulated qualitative snapshots using $l_0 = 0.002$ m are presented in Figure 16. As shown in Figure 16A,B, the pressure peak takes place at the bottom corner near the inlet. Then the pressure gradually decreases along the spreading direction. Namely, pressure gradients provide the driving forces for the spreading. On the other hand, the viscosity forces are drag forces to slow down the flow. Clear velocity gradient can be observed near the substrate, as shown in Figure 16C,D, which indicates that the spreading is in the gravity-viscosity regime. Namely, Equation (56) can provide reliable prediction. From Figure 16, the spreading flow simulated by the existing method is slower than that by the proposed method. The quantitative results will be discussed below for comparing the two methods.

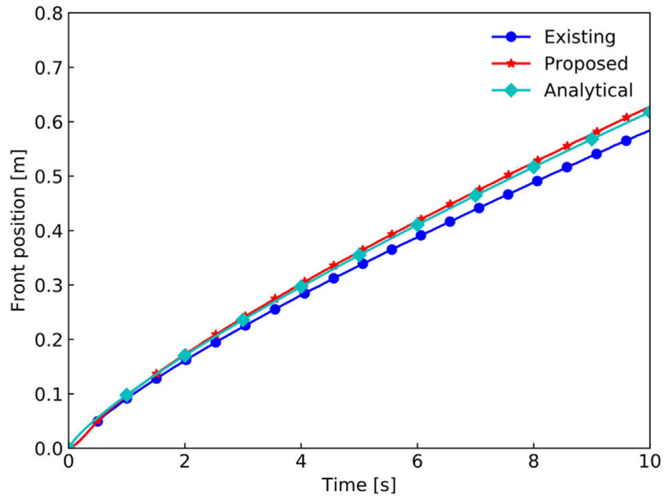
The front position histories simulated by the two methods with different resolutions are compared in Figure 17. When the particle size is coarse (ie, 0.004 m), the front position history predicted by the proposed method is obviously more accurate than that by the existing method (Figure 17A), when compared with the analytical result. When fine particles are employed (Figure 17B,C), the difference between the results by the two methods becomes small. This indicates that both methods exhibit convergence. However, the proposed method converges faster than the existing one. Briefly, the proposed method is more accurate that the existing one in terms of predicting front positions.

The computational time information is summarized in Table 1. The relative increase of computational cost is around 3%-13.8%. Considering that in dam break flow is 13.9%, it can be generally said that the increase of computational cost is around 10%-15%.

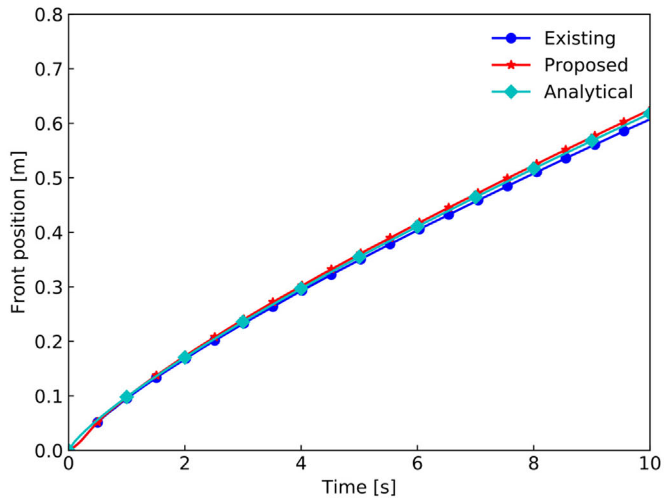
In summary, the improved velocity boundary conditions in the new schemes can help to provide more accurate results than the existing method in the such laminar free surface flow.



(A) $l_0 = 0.004$ m



(B) $l_0 = 0.002$ m



(C) $l_0 = 0.001$ m

FIGURE 17 Comparisons of simulated and analytical front position histories using different resolutions [Color figure can be viewed at wileyonlinelibrary.com]

Computational time (s)	$l_0 = 0.004$ m	$l_0 = 0.002$ m	$l_0 = 0.001$ m
Existing method	35.98	273.45	3997.24
Proposed method	37.18	281.46	4550.18
Rate of increase (%)	3.34	2.93	13.8

TABLE 1 The computational time in simulations of the viscous spreading flow

5 | CONCLUDING REMARKS

In this article, the corrective matrix schemes were newly generalized to incorporate the straightforward treatment of Neumann boundary condition by redefining the directional difference (d_{ij}) and generalized direction vector ($\boldsymbol{P}$). When the high-order schemes were directly applied to free surface particles, instability could easily take place mainly because of the biased error caused by the biased neighbor support, large interaction coefficients caused by less neighbor particles via corrective matrix and free surface boundary fluctuations caused by the surface particle detection methods. The existing stable schemes with virtual particles in References 31,32 were applied to free surface and nearby particles to provide a stable transitional layer for free surface flow. The new schemes were applied to the internal particles under the stable transitional layer. In other words, the wall boundary condition treatment methods for velocity and pressure are improved in the proposed method compared to the existing one.[47] The numerical examples of free surface flows demonstrated the proposed method could improve the accuracy and the stability such as spurious fluctuations of pressure and velocity in comparison to those of the existing one. The computational cost increased around 10%-15% relative to the existing method.

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